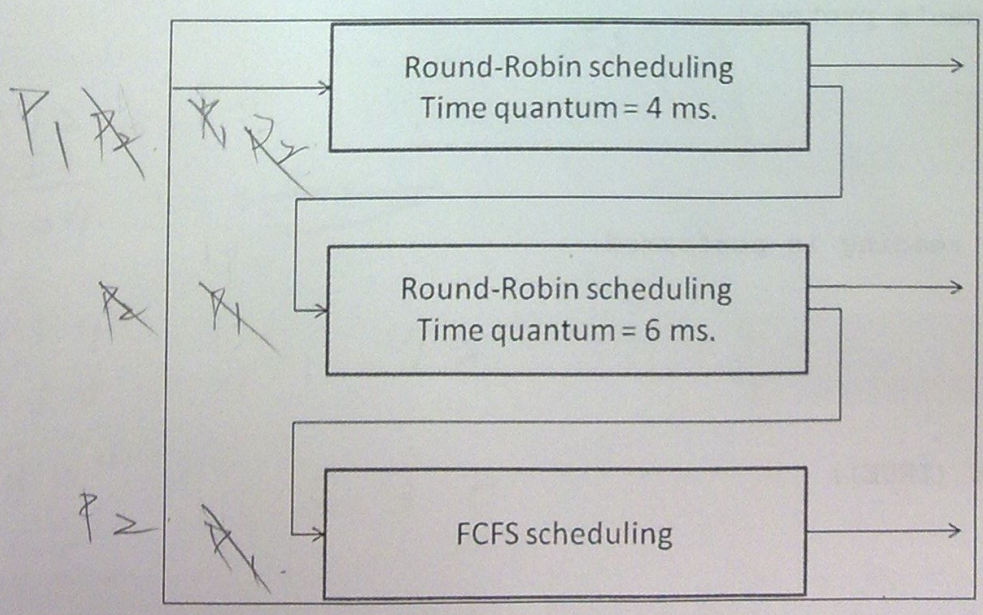
Operating Systems – Final Examination
June 19, 2012
Total: 100 points

1. Explain the following terms:
   (a) (2 points) Processor affinity
   (b) (2 points) Critical section
   (c) (2 points) Safe sequence of processes
   (d) (2 points) The worst-fit strategy of contiguous memory allocation
   (e) (2 points) Inverted page table

2. Scheduling – Consider the following two processes P_1 and P_2, each of which consists of two CPU bursts and one I/O burst. Assume that P_1 and P_2 arrive at time **0** and **2** (millisecond) respectively:

P_1:	CPU burst: 8 milliseconds.	I/O burst: 4 milliseconds.	CPU burst: 8 milliseconds.
P_2:	CPU burst: 2 milliseconds.	I/O burst: 2 milliseconds.	CPU burst: 16 milliseconds.

   (a) (10 points) Draw a Gantt chart illustrating the execution of the processes using the following **preemptive multilevel feedback-queue scheduling algorithm**. **Note that a high-level process will preempt a low-level process and a process will be scheduled to a lower level if it is preempted**.



   (b) (10 points) What is the average waiting time for the scheduling algorithm?

3. Synchronization

(a) (10 points) Complete the following entry and exit code fragments of Peterson's solution for process P_0.

(b) (10 points) show that the code fragments satisfy the three requirements of critical section solution.

```
// shared data for Peterson's solution
boolean flag[2];
int turn = 1;

do {  flag[0] = true;

      while (turn == 1 && flag[1]);

      CRITICAL SECTION

      flag[0] = false;

      REMAINDER SECTION

} while (TRUE);
```

mutex

progress

4. (10 points) In the first readers-writers problem, readers have higher priorities to access a shared object (i.e., file). Design the entry and exit sections of the reader processes using the following (shared) data structures:

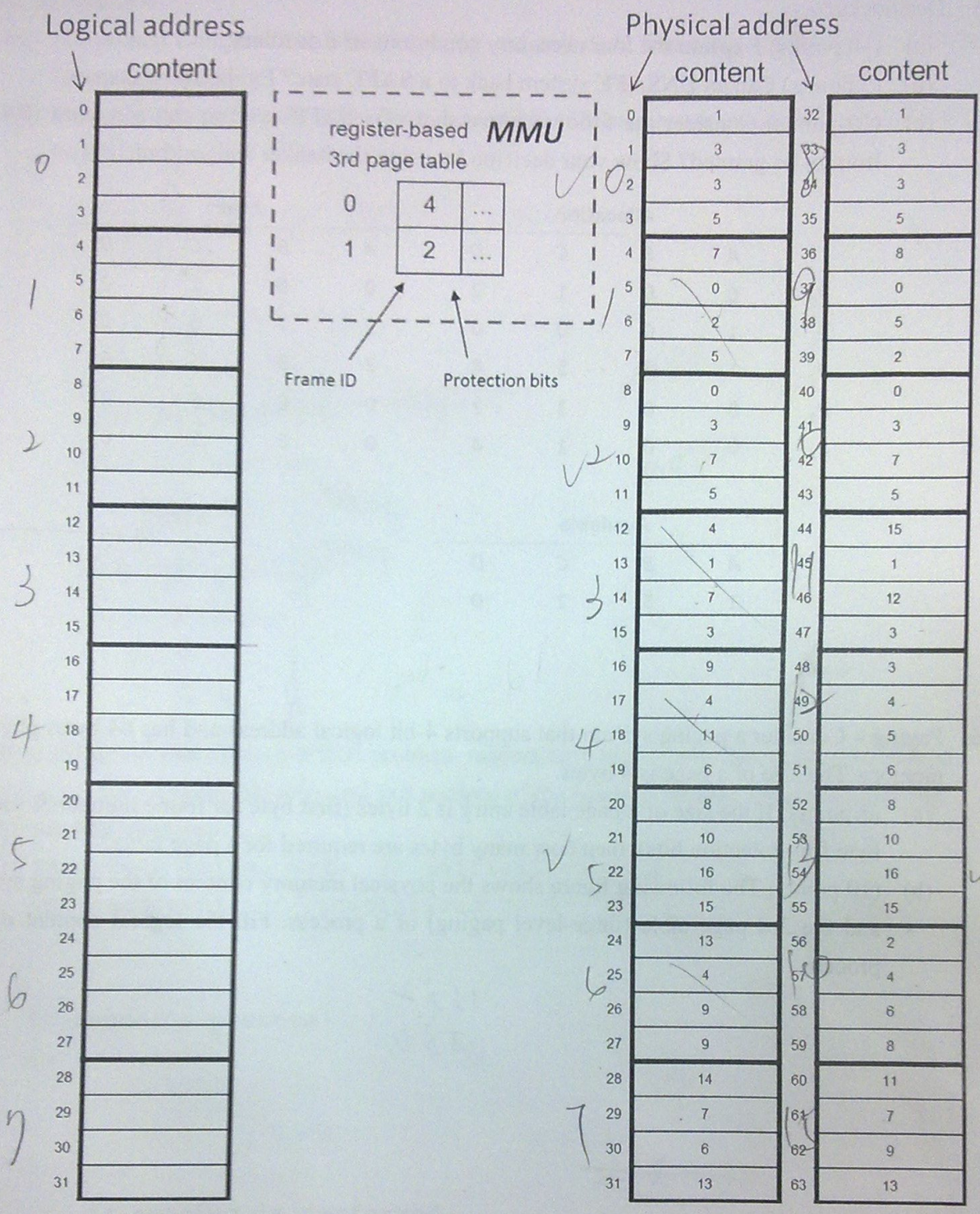
```
semaphore mutex, wrt;  // initialized to 1
int readcount;  // initialized to 0


// reader's protocal
do {


    // reading is performed


} while (TRUE);
```

5. Deadlocks:
   (a) (10 points) **Explain** the four necessary conditions of a deadlock.
   (b) (5 points) Can an **UNSAFE** system back to a **SAFE** state? Explain you answer.
   (c) (10 points) Consider the following snapshot of a SAFE system; can a request (0,4,2,0) from p_1 be granted? Show your decision by using the banker's algorithm.

	Allocation				Max			
	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2
P_1	1	0	0	0	1	7	5	0
P_2	1	3	5	4	2	3	5	6
P_3	0	6	3	2	0	6	5	2
P_4	0	0	1	4	0	6	5	6

Available			
A	B	C	D
1	5	2	0

6. Paging – Consider a paging system that supports 4-bit logical address and has 64 bytes physical memory. The size of a page is 4 bytes.
   (a) (5 points) If the size of a page table entry is 2 bytes (first byte for frame number & second byte for protection bits), then how many bytes are required for a page table?
   (b) (10 points) The following figure shows the physical memory content of the paging system and the 3rd page table (three-level paging) of a process. Fill the logical content of the process.


Logical address
content
Physical address
content
content
register-based MMU
3rd page table
0
1
4
...
2
...
Frame ID
Protection bits
0
1
2
3
4
5
6
7
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
1 3 3 5 7 0 2 5 0 3 7 5 4 1 7 3 9 4 11 6 8 10 16 15 13 4 9 9 14 7 6 13
32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63
1 3 3 5 8 0 5 2 0 3 7 5 15 1 12 3 3 4 5 6 8 10 16 15 2 4 6 8 11 7 9 13